

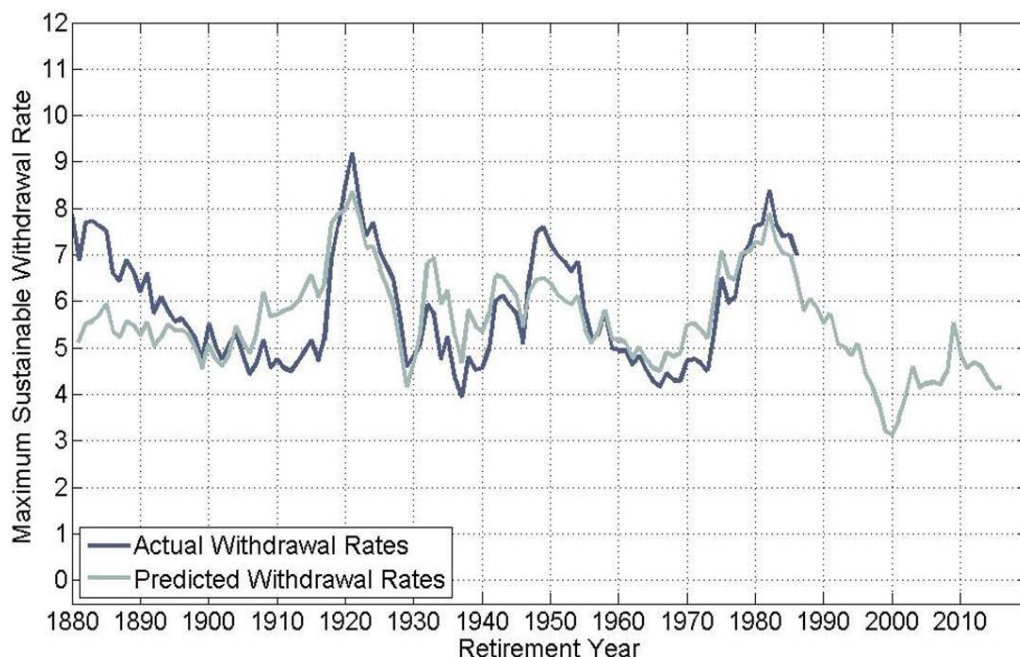
article “Valuation Ratios and the Long-Run Stock Market Outlook.” They found useful predictive power in estimating the relationship between Shiller’s PE10 and the subsequent ten-year real return for stocks. Returns tend to be lower after high valuation periods and vice versa.

Exhibit 5.8

Actual and Predicted Maximum Sustainable Withdrawal Rates

For 50/50 Asset Allocation, 30-Year Retirement Period

Estimated for Years 1881–1987, Explanatory Variables: PE10 and 10-Year Treasury Rate



Sustainable withdrawal rates from a diversified portfolio including stocks can be expected to share this relationship with market valuations. Using a statistical method called “multivariate regression analysis,” we can analyze the relationship between the maximum sustainable withdrawal rate over a thirty-year retirement with a 50/50 asset allocation to stocks and bonds, and the values of Shiller’s PE10 and the ten-year Treasury rate at the start of those retirements. The statistical analysis results in this equation:

$$\text{Predicted Withdrawal Rate} = 12.15 - 2.47 \times \log(\text{PE10}) + 0.05 \times \text{Bond Yield}$$